Subspaces

4.1 Column Space, Row Space, and Nullspace

Column and Row Space: Definitions

Let A be an $m \times n$ matrix, represented as:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

1. **Row Space of A**:
   - The subspace of $\mathbb{R}^n$ spanned by the rows of A. This means it consists of all possible linear combinations of the row vectors of A.
   - Notation: $\text{Row}(A)$.
   - $\text{Row}(A) = \text{span}\{(a_{11}, a_{12}, \ldots, a_{1n}), (a_{21}, a_{22}, \ldots, a_{2n}), \ldots, (a_{m1}, a_{m2}, \ldots, a_{mn})\}$.
2. **Column Space of A**:
   - The subspace of $\mathbb{R}^m$ spanned by the columns of A. This means it consists of all possible linear combinations of the column vectors of A.
   - Notation: $\text{Col}(A)$.
   - $\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}, \ldots, \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} \right\}$.

Example

Let
$A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix}$

- **Column Space of A**

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} \right\} = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

- **Row Space of A**
  $\text{Row}(A) = \text{span}\{(1, 0, 2, 0), (0, 1, 0, 2), (1, 1, 2, 2)\} = \text{span}\{(1, 0, 2, 0), (0, 1, 0, 2)\}$

Important Note

Column vectors are represented as vectors in $\mathbb{R}^m$, while row vectors (after transposition) are represented as vectors in $\mathbb{R}^n$.

Row Operations and Row Space

Row operations do not change the row space of a matrix. This is because row operations produce linear combinations of existing rows, so the new rows are still within the span of the original rows. Also, row operations are reversible, meaning the original rows are within the span of the new rows.

Theorem: Row Operations Preserve Row Space

Suppose A and B are row equivalent matrices. Then the row space of A is equal to the row space of B, i.e., Row$(A) = \$Row(B).

Sketch of Proof:

1. *Rearrangement of Rows*: If B is obtained from A by simply interchanging rows, then the set of rows remains the same, and hence the span (row space) is unchanged.
2. *Scalar Multiplication*: If a row is multiplied by a nonzero scalar, it still lies in the same span.
3. *Adding a Multiple of One Row to Another*: If we replace a row with the sum of that row and a scalar multiple of another row, the new row is still in the span of the original rows.

Finding a Basis for the Row Space

Theorem: If a matrix R is in reduced row-echelon form, then the nonzero rows of R form a basis for its row space.

Algorithm

1. Reduce the matrix A to its reduced row-echelon form R.
2. Identify the nonzero rows in R.
3. These nonzero rows form a basis for the row space of A.

Example

Let $A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix}$. The reduced row echelon form is $R = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

A basis for Row(A) is {(1 0 2 0), (0 1 0 2)}.

Finding a Basis for the Column Space

Theorem: Given a matrix A, transform it to its reduced row-echelon form R. The columns of A corresponding to the pivot columns in R form a basis for the column space of A.

Algorithm

1. Reduce the matrix A to its reduced row-echelon form R.
2. Identify the pivot columns in R (the columns with leading 1s).
3. Select the corresponding columns from the *original* matrix A.
4. These columns form a basis for the column space of A.

Important Caveat

Remember to select the basis vectors from the original matrix A, not from the reduced form R.

Example

Starting from the previous matrix $A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix}$ with RREF $R = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. The pivot columns are the first and second columns. Thus, a basis for the column space of A is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Key Ideas

- RREF is only a tool. Remember to go back to the original matrix when extracting column space basis vectors.
- Linear relations between columns are preserved in RREF.

Nullspace

- Definition: the nullspace of an $m \times n$ matrix A is the solution space to the homogeneous system $Ax = 0$. $Null(A) = v \in R^n | Av = 0$

4.2 Rank

Question

For any matrix A, is the dimension of the column space of A equal to the dimension of the row space of A? True or false?

Definition: Rank

Let A be an $m \times n$ matrix, and let R be its reduced row-echelon form.

- $\dim(\text{Col}(A)) =$ the number of pivot columns in R.
- $\dim(\text{Row}(A)) =$ the number of nonzero rows in R.

The *rank* of A is defined to be the dimension of its column or row space:
rank$(A) =$ dim(Col(A)) $=$ dim($Row(A)$).

rank(A) and rref(A)

The rank of A is equal to the number of non zero rows in rref(A).

Exercise

Prove that the rank is invariant under transpose, rank$(A) =$ \$$rank$(A^T)\$.

Example

A = $\begin{pmatrix} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

Transform to rref(A)= $\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$

rank(A) + nullity(A) = 2 + 2 = 4 = number of columns of A. Indeed,
$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \right\}$, is a basis of the column space, and $\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$, is a basis for the nullspace.

Observe that the column space is a subspace of R^3 but the subspace is a subspace of R^4.

Rank-Nullity Theorem

Theorem: (Rank-Nullity Theorem) Let A be an $m \times n$ matrix. The sum of its rank and nullity is equal to the number of columns:
$\text{rank}(A) + \text{nullity}(A) = n.$

Proof Sketch:
The nullity of A is equal to the number of non-pivot columns in its reduced row-echelon form, and the rank of A is equal to the number of pivot columns in its reduced row-echelon form.

Definition: Full Rank

An $m \times n$ matrix A is said to be of *full rank* if its rank is equal to the minimum of the number of rows or columns:

$\text{rank}(A) = \min\{m, n\}$

Discussion: Full Rank and Square Matrices

If A is a square matrix $(m = n)$, then A is full rank if and only if it is invertible. This ties together many concepts:

- The rows and columns of A are linearly independent.
- The homogeneous system $Ax = 0$ has only the trivial solution.
- A is expressible as a product of elementary matrices.
- A has a left and a right inverse

Theorem: Full Rank and Number of Columns

Suppose A is a $m \times n$ matrix. The following are equivalent:

1. A is full rank, where the rank is equal to the number of columns, rank(A) = n.
2. The rows of A spans R^n, Row(A) = R^n.
3. The columns of A are linearly independent.
4. The homogenous system Ax = 0 has only the trivial solution, that is, Null(A) = {0}.
5. A^T A is an invertible matrix of order n.
6. A has a left inverse.

If rank(A) = m, then the reduced row-echelon form is of the form

$$\begin{pmatrix} 1 & 0 & \dots & 0 & \dots \\ 0 & 1 & \dots & 0 & \dots \\ 0 & 0 & \dots & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \end{pmatrix}$$

Theorem: Full Rank and Number of Rows

Suppose A is a $m \times n$ matrix. The following are equivalent:

1. A is full rank, where the rank is equal to the number of rows, rank(A) = m.
2. The columns of A spans R^m, Col(A) = R^m.
3. The rows of A are linearly independent.
4. The linear system Ax = b is consistent for every b ∊ R^m.
5. AA^T is an invertible matrix of order m.
6. A has a right inverse.

If rank(A) = m, then the reduced row-echelon form is of the form

$$\begin{pmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ 0 & 0 & \dots & 1 & \dots & 0 \end{pmatrix}$$

Caution notes:

Row operations do not perserve column space.

Summary of Subspaces Associated with a Matrix

Let A be an $m \times n$ matrix.

Subspace	Subspace of	Basis	Dimension
Col(A)	Rm	Columns of A corresponding to pivot columns in RREF	rank(A) = no. of pivot columns in RREF
Row(A)	Rn	Nonzero rows of RREF	rank(A) = no. of nonzero rows in RREF
Null(A)	Rn	Vectors in general solution to Ax = 0	nullity(A) = no. of nonpivot columns in RREF